Research Foundation: Week 1

Remote Patient Health Monitoring Using Machine Learning

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1. Research question and working hypothesis

How do Gaussian Naive Bayes and Random Forest compare in classifying selected MIT-BIH ECG heartbeat classes under a subject-disjoint evaluation design? I expect that the algorithms may differ in macro-F1 and class-specific precision and recall, but I have not established a direction of superiority. I will finalize the classes, exclusions, and features before training. This is an offline comparison, not a live monitoring application, heart-attack predictor, or clinically validated diagnostic system.

2. Intended research evidence

I will compare held-out predictions, class supports, macro-F1, per-class precision/recall/F1, and confusion matrices against a majority-class reference. Models will use identical evaluation partitions and consistent preparation. The evidence chain is: research question -> annotated ECG recordings -> validated preparation and subject-disjoint evaluation -> classification errors -> bounded comparative claims. Week 1 establishes input accessibility, structure, numerical completeness, and annotation distribution. These descriptive results do not support or reject the classifier hypothesis.

3. Research inputs

I acquired records 100 and 106 from the MIT-BIH Arrhythmia Database, version 1.0.0, using versioned PhysioNet URLs. Each record includes header (.hea), signal (.dat), and reference annotation (.atr) files. My download manifest records six downloads, byte sizes, and SHA-256 hashes. The hashes identify the local files; I have not compared them against provider checksums. The records are a convenience pilot, not a representative clinical sample. I will retain acquisition instructions rather than commit raw ECG files. Original annotation symbols are preserved pending a target-class policy.

4. Computational environment and code organization

I executed the notebook locally in Jupyter using the Python (ECG Research) kernel. My execution environment used Python 3.11.18, NumPy 2.4.6, pandas 2.3.3, Matplotlib 3.11.2, and WFDB 4.3.1. The primary computational artifact is src/week1_ecg_research_foundation.ipynb. The principal dependency list and environment snapshot belong under src/. Generated tables, plots, manifests, and execution records belong under results/week_1/. This report belongs under docs/step_2/week_1/. The environment snapshot records my local configuration; it does not guarantee installation on every platform. The notebook, rather than a standalone experiment script, is the execution procedure.

5. Execution instructions

Use Python 3.11 with Jupyter and an appropriate ipykernel. Install the principal packages with python -m pip install -r src/requirements.txt. Select the configured kernel, open the notebook, and run cells in order: imports/configuration, acquisition, loading and validation, numerical summaries, annotation counts, visualization, and execution export. Verify the displayed project root. When the working directory is src/, the code uses its parent; otherwise it uses the current directory. Internet is needed for initial downloads.

Existing nonempty files are reused. Skip the optional installation cell when dependencies are installed. Save the notebook with outputs. My saved outputs document sequential interactive execution; a clean-kernel full rerun remains a reproducibility check to complete.

6. Completed computation and preliminary results

I obtained and loaded both records. Each contains 650,000 samples per channel, two channels, and a 360-Hz sampling frequency, corresponding to approximately 30.09 minutes. Record 100 contains MLII/V5 and 2,274 annotation entries; record 106 contains MLII/V1 and 2,098 entries. All 14 structural checks passed, covering signal/header dimensions, channel metadata, annotation lengths, integer indices, range, and ordering. Summed symbol counts matched loaded entries. All four channels contained zero NaN and infinite values. MLII standard deviations were approximately 0.1932 mV and 0.3681 mV for records 100 and 106. I did not filter, normalize, or impute the signals. I generated ten-second MLII plots with reference labels and a logarithmic annotation-count chart. All eight expected inspection outputs were present and nonempty. File presence is not independent scientific validation. My execution summary records that no model was trained and no classification metrics were computed.

7. Problems, limitations, and design implications

My initial installation in system-wide Anaconda failed with a Windows access-denied error. I resolved the imports by creating a user-directory Conda environment and registering a dedicated Jupyter kernel. Numerical completeness does not rule out noise or artifacts. Different second leads require a consistent channel policy. Record 100 contains N=2,239, A=33, V=1, and +=1; record 106 contains N=1,507, V=520, +=41, and ~=30. Non-beat events must not automatically become target heartbeat classes. The distributions show limited class coverage and imbalance, not population prevalence. Two records are insufficient for a defensible generalization comparison. I will define subject groups before splitting and keep repeated records from one subject together. I have not implemented segmentation, feature extraction, data partitions, classifiers, or clinical validation. Reviewing XDTEncoder remains part of the scholarly context; I have not reproduced its architecture.

8. Planned Week 2 work

I will finalize target symbols, exclusions, channel selection, and adequate records; map records to subjects; and define disjoint evaluation partitions. I will implement beat windows with boundary checks and shared features. Data-dependent preprocessing and any resampling will be fitted on training data only. I will implement a majority-class reference and Gaussian Naive Bayes, followed by Random Forest using the same conditions. I will save parameters, seeds, predictions, class supports, macro-F1, per-class metrics, and confusion matrices, and add checks for leakage and feature-label alignment. LSTM remains optional. Any design revision will be justified methodologically rather than chosen to force an expected result.

References and supporting files

Moody, G., & Mark, R. (2005). MIT-BIH Arrhythmia Database (version 1.0.0) [Data set]. PhysioNet. https://doi.org/10.13026/C2F305
WFDB developers. (n.d.). Input/output documentation. https://wfdb.readthedocs.io/en/latest/io.html
Supporting files: executed notebook; record_metadata.csv; validation_checks.csv; signal_quality_summary.csv; annotation_counts.csv; download_manifest.json; environment.json; execution_summary.json; output_file_checks.csv; and generated signal and annotation plots.